ON A NODAL CURVE WITHOUT RATIONAL TAILS, EVERY PROPER REDUCED CONNECTED SUBSCHEME HAS NONNEGATIVE CANONICAL DEGREE AT LEAST ITS FIRST COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the restriction sequence with the degree identity, the divisibility argument isolating the exceptional configuration, the irreducible-case conclusion, and the reducible-case component reduction are all retained verbatim.

Editorial note. The label lemma-no-rational-tail names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The subscheme-complement duality package. Same chapter, Lemmas lemma-closed-subscheme-reduced-gorenstein, lemma-facts-about-nodal-curves and lemma-closed-subscheme-nodal-curve. Supplies the complement Z, the restriction sequence 0 -> omega_Y -> omega_X|_Y -> O_{Y cap Z} -> 0, and the nodal bookkeeping used throughout.
- Divisibility of numerical invariants. Varieties, Lemma varieties-lemma-divisible. Makes [k' : k] divide all numerical invariants of Y over k' = H^0(Y, O_Y), forcing failure cases to be defined over k'.
- Rational tails and component degrees. Same chapter, Example example-rational-tail; Varieties, Lemma varieties-lemma-degree-in-terms-of-components. Identifies the exceptional configuration with a rational tail and compares component degrees in the reducible case.

Source reference. Original reference 0E3H: sources/stacks-curves/pinned-original-source.tex, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 6036; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E38: sources/stacks-curves/pinned-original-source.tex, source lines 4794–4794. Prerequisite pointer invoked at proof line(s) 6052; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0E34: sources/stacks-curves/pinned-original-source.tex, source lines 1012–1012. Prerequisite pointer invoked at proof line(s) 6049; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E37: sources/stacks-curves/pinned-original-source.tex, source lines 4763–4763. Prerequisite pointer invoked at proof line(s) 6051; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYW: sources/stacks-chow/tags.txt, source lines 13804–13804. Prerequisite pointer invoked at proof line(s) 6074; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C6P: sources/stacks-chow/tags.txt, source lines 15304–15304. Prerequisite pointer invoked at proof line(s) 6047; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Assume the singularities of X are at-worst-nodal. If X does not have a rational tail (Example 0E3H), then for every reduced connected closed subscheme $Y \subset X$, $Y \neq X$ of dimension 1 we have $\deg(\omega_X|_Y) \geq \dim_k H^1(Y, \mathcal{O}_Y)$.*

Proof. Let $Y \subset X$ be as in the statement. Then $k' = H^0(Y, \mathcal{O}_Y)$ is a field and a finite extension of k and $[k' : k]$ divides all numerical invariants below associated to Y and coherent sheaves on Y, see Varieties, Lemma 0C6P. Let $Z \subset X$ be as in Lemma 0E34. We will use the results of this lemma and of Lemmas 0E37 and 0E38 without further mention. Then we get a short exact sequence

$$0 \to \omega_Y \to \omega_X|_Y \to \mathcal{O}_{Y \cap Z} \to 0$$

See Lemma 0E34. We conclude that

$$\deg(\omega_X|_Y) = \deg(Y \cap Z) + \deg(\omega_Y) = \deg(Y \cap Z) - 2\chi(Y, \mathcal{O}_Y)$$

Hence, if the lemma is false, then

$$2[k' : k] > \deg(Y \cap Z) + \dim_k H^1(Y, \mathcal{O}_Y)$$

Since $Y \cap Z$ is nonempty and by the divisiblity mentioned above, this can happen only if $Y \cap Z$ is a single k'-rational point of the smooth locus of Y and $H^1(Y, \mathcal{O}_Y) = 0$. If Y is irreducible, then this implies Y is a rational tail. If Y is reducible, then since $\deg(\omega_X|_Y) = -[k' : k]$ we find there is some irreducible component C of Y such that $\deg(\omega_X|_C) < 0$, see Varieties, Lemma 0AYW. Then the analysis above applied to C gives that C is a rational tail. □